

Examiner
only

5. Some heterotrophic organisms are also saprotrophic. Fungi are examples of saprotrophic organisms.

(a) (i) State the meaning of the term heterotrophic organism. [1]

.....

.....

.....

(ii) Describe the process of saprotrophic nutrition in fungi. [3]

.....

.....

.....

.....

.....

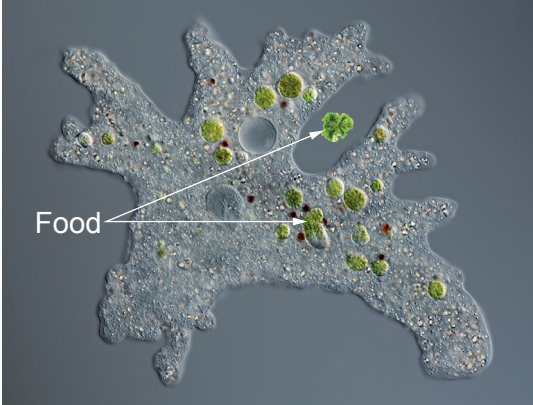
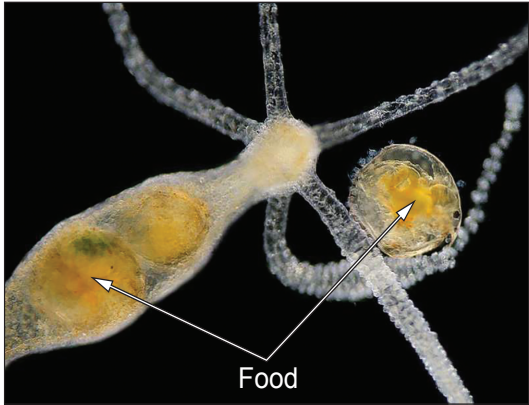
.....

.....



Examiner
only

- (b) *Amoeba* and *Hydra* are also heterotrophic organisms. **Images 5.1** and **5.2** show *Amoeba* and *Hydra* feeding. The images also show food which has previously been taken into the organisms.

Image 5.1 <i>Amoeba</i> feeding	Image 5.2 <i>Hydra</i> feeding
	

Use **Images 5.1, 5.2** and your knowledge of digestion to describe **one** similarity and **two** differences between the adaptations of *Amoeba* and *Hydra* for obtaining their nutrition.

[3]

Similarity:

.....

.....

Difference I:

.....

.....

Difference II:

.....

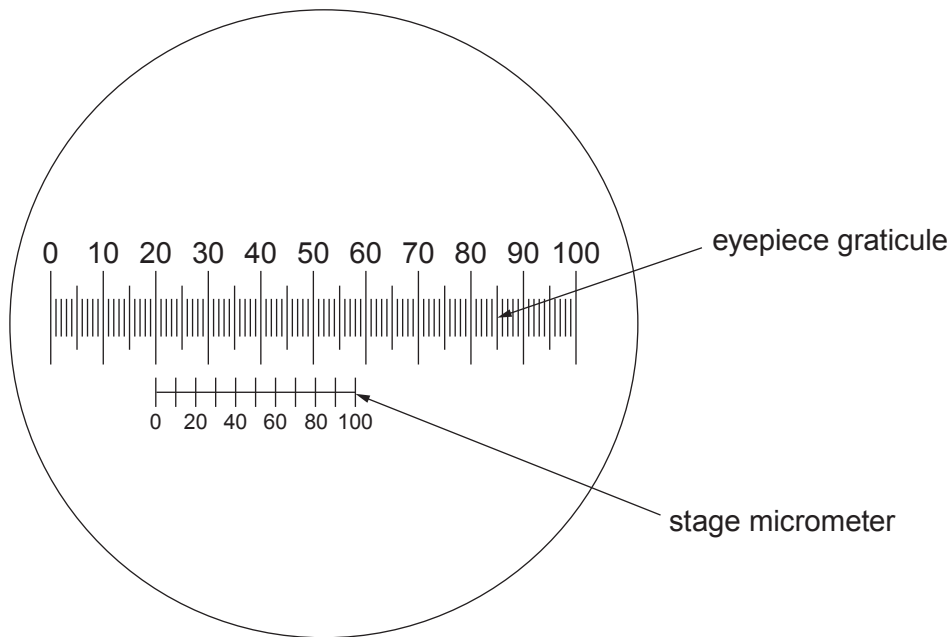
.....



- (c) A student used an eyepiece graticule to measure the length of an *Amoeba* on a microscope slide at a magnification of $\times 40$. In order to calculate the actual size of the *Amoeba*, the student had to calibrate the eyepiece graticule. She used a 1 mm stage micrometer slide which had 100 divisions.

Image 5.3 shows the field of view from a microscope with the eyepiece graticule and stage micrometer labelled.

Image 5.3



- (i) Use **Image 5.3** to calculate the length of **one** eyepiece unit in micrometres. [2]

One eyepiece unit = μm

- (ii) The *Amoeba* at $\times 40$ magnification measured 27 eyepiece units (epu). Calculate the actual length of the *Amoeba* in micrometres. [1]

Amoeba length = μm

10

